

Grace Wednesday warm-leads: (1) $\eta(1405)$ at rung 9 — CLOSED NEGATIVE (coincidence; it sits far below the BST 0^{-+} glueball and is a $q\bar{q}$ state, no substrate 0^{-+} at seat 9). (2)

Operator-algebraic-closure sweep (Casey Wed-AM question): does it drive CKM/PMNS? Answer: NO — CKM/PMNS is the ‘overlap’ driver, distinct from the gauge-sector ‘closure’ driver. Produces the BST MULTIPLICITY-DRIVER MAP: four drivers (closure / Pauli / strata / overlap), one per operator class, unified under one meta-principle. Correctly scopes the K498 operator-closure mechanism as gauge-sector-specific.

Grace

2026-06-24 Wednesday

BST multiplicity-driver map + two warm-lead verdicts

Lead 1 — $\eta(1405)$ at rung 9: CLOSED NEGATIVE

$\eta(1405) = 1409 \text{ MeV} = \text{seat } 9.01$ (0.1% from rung 9). But the BST 0^{-+} glueball ground sits at $\sim 2580 \text{ MeV}$ (seat 16.5) in the Lyra ladder — $\eta(1405)$ is *far below* it and cannot be the lightest pseudoscalar glueball. Seat 9 is a discrete-series rung ($\frac{1}{2}C_2=9$), but no 0^{-+} glueball sits there in the ladder. Phenomenologically $\eta(1405)$ is a $q\bar{q}$ pseudoscalar (η radial excitation / the $\eta(1405)$ - $\eta(1475)$ mixing puzzle), not a glueball. Verdict: coincidence, look-elsewhere tail, do NOT bank. Lead closed-negative. (Honest result — the mechanism check returns no substrate state at seat 9, so the Tuesday mass-coincidence stays a coincidence.)

Lead 2 — operator-algebraic-closure sweep $\rightarrow$ the multiplicity-driver map

Casey’s Wednesday question (via K498): does the operator-algebra-closure mechanism that forces the four glueball channels also drive CKM/PMNS multiplicity? Sweeping the BST corpus, the answer is no — but it places the mechanism in a clean map. BST generates multiplicity by four distinct drivers, one per operator class:

operator class	driver	mechanism	multiplicity	count source
gauge tensor $F \otimes F$	operator-algebra CLOSURE	decompose ONE tensor into irreducibles	4 J^{PC} channels $\{0^{++}, 0^{-+}, 2^{++}, 1^{+-}\}$	4D Lorentz decomposition
fermion content	Pauli antisymmetry	shell filling	nuclear magic 28/50/82/126	shell capacities

operator class	driver	mechanism	multiplicity	count source
matter generations	support-orbit STRATA	D_IV ⁵ has rank+1 = 3 strata (Korányi-Wolf, F86)	3 generations	rank + 1
mixing CKM/PMNS	OVERLAP	misalignment of two non-commuting mass operators across the 3 strata (F84 Bergman kernel $\times 3$)	3 $\times$ 3 mixing	strata overlaps

The key distinction (why closure $\neq$ CKM/PMNS)

- Glueball multiplicity = the irreducible decomposition of ONE tensor ($F \otimes F$). Closure forces all four components to have eigenstates. Count (4) comes from the 4D Lorentz structure.
- CKM/PMNS multiplicity = the misalignment of TWO operators (Yukawa_{up} vs Yukawa_{down}) across the 3 support strata. Count (3) comes from rank+1 strata, not a tensor decomposition.

These are different operator-algebraic mechanisms — one-tensor-decomposition vs two-operator-misalignment. So the gauge-sector closure mechanism does not transfer to the matter/mixing sector.

The meta-principle that DOES unify (honest tier)

The substrate supports the full dimensional content of its constituent operators; the *driver* depends on the operator class. Gauge tensor $\rightarrow$ closure; fermion $\rightarrow$ Pauli; matter $\rightarrow$ strata; mixing $\rightarrow$ overlap.

One meta-principle, four drivers. This correctly scopes the K498 operator-closure mechanism: it is the gauge-sector instance of the meta-principle, *not* a universal multiplicity law (consistent with K498’s own “NOT saying this is universal” caveat). For Paper A §7, the operator-closure mechanism should be presented as the gauge-sector driver, with the meta-principle as the broader frame and CKM/PMNS explicitly a *different* driver — that keeps the “WHY four channels” answer honest and bounded rather than over-claiming a universal principle.

Net (warm-leads slot)

- $\eta(1405)$: closed-negative (coincidence).
- Operator-closure sweep: does not extend to CKM/PMNS (different driver). Produced the multiplicity-driver map (4 drivers, 1 meta-principle) — a curation artifact that scopes the Wed-AM mechanism correctly for Paper A §7. No new banked claim; the value is the honest scoping + the map.

Connections: K498 (operator-closure, gauge driver), F86 (strata $\rightarrow$ generations), F84 (Bergman overlaps $\rightarrow$ mixing), Mayer-Jensen (Pauli $\rightarrow$ shells), T2490 (primaries = the substrate’s operator-algebra spectrum bottom — the closure made visible at primary scale). Count HOLDS 4 of 26.

— Grace, 2026-06-24 Wednesday (warm-leads slot).